ORB Feature Extraction and Frame-to-Frame Matching on TUM VI (room1)

1. Objective

Extract ORB features from the TUM VI dataset camera images and match them between consecutive frames, as a first stage toward visual odometry (VO) and visual-inertial odometry (VIO).

2. Data

- Dataset: TUM VI, `dataset-room1_512_16.bag` (calibrated, 512x512, 16-bit grayscale).
- Topic used: `/cam0/image_raw` (left camera), 2821 frames.
- Earlier runs on calibration sequences (`calib-cam7`, `calib-cam4`) were used only to validate the code, not to report results.

3. Pipeline

1. Read the bag with `rosbags` (no ROS installation needed).
2. Decode each message into a 512x512 16-bit array.
3. Convert to 8-bit with a fixed scale (divide by 257), so the same physical point keeps the same brightness across frames.
4. Detect ORB keypoints and compute 256-bit binary descriptors (OpenCV).
5. Match each frame to the previous one with a brute-force matcher using Hamming distance (k = 2).
6. Ratio test: keep a match only if its best distance is below 0.75 times the second best.
7. RANSAC on the fundamental matrix to remove geometrically inconsistent matches (inliers).
8. Save per-frame statistics (CSV), plots and match visualizations.

4. Final parameters

Parameter	Value
Features per frame	1000
FAST threshold	10
Ratio test	0.75
RANSAC threshold / confidence	1.0 px / 0.99
16-bit to 8-bit conversion	divide by 257 (fixed)

5. Results (room1, 2821 frames, FAST threshold 10)

Metric	Value
Mean ratio-test matches per frame	555.8
Mean RANSAC inliers per frame	386.3
Inlier share (inliers / ratio-test matches)	69.5%
Keypoints per frame (min / mean / max)	268 / 966.3 / 1000
Frames with fewer than 400 keypoints	5
Frames with fewer than 100 inliers	5

Because the mean keypoint count is close to the 1000-feature cap, inliers per frame is used as the main metric.

6. Parameter tuning: FAST threshold 20 vs 10

Metric	Threshold 20 (default)	Threshold 10 (final)
Mean ratio-test matches per frame	498.4	555.8
Mean RANSAC inliers per frame	353.1	386.3
Inlier share	70.8%	69.5%

Metric	Threshold 20 (default)	Threshold 10 (final)
Keypoints per frame (min / mean / max)	121 / 847.5 / 1000	268 / 966.3 / 1000
Frames below 400 keypoints	125	5
Frames below 100 inliers	41	5

Lowering the threshold raised mean inliers by about 9% and almost removed the frames with very few keypoints, at a cost of 1.3 points of inlier share.

7. Weak-frame analysis

At threshold 20, 125 frames (4.4%) had fewer than 400 keypoints, usually in short runs of about five frames. In a detailed window (frames 880-924) the following was observed:

- Mean image brightness changes in steps, and keypoint counts shifted by roughly 70-100 at those steps.
- Weak frames occurred at the end of a low-brightness plateau where image sharpness (Laplacian variance) was also lowest.
- A brighter but less sharp stretch had about 46% fewer keypoints than a sharper one, so sharpness or texture affects keypoint count in addition to exposure.
- On the dark plateau (frames 894-902), threshold 10 raised mean keypoints from 421 to 823 and inliers per frame from 182 to 308, at a cost of 3 points of inlier share.

The measurements do not separate motion blur from reduced scene texture as the cause.

8. Limitations

- RANSAC inliers are matches consistent with one camera motion, not verified correct matches. No ground-truth comparison was made.
- The fundamental matrix is unreliable under pure camera rotation.
- The calibration sequences were used for code validation only.
- Results come from a single sequence and one parameter sweep on a 45-frame window plus the full runs.

9. Next steps

- Compare estimated motion against the motion-capture ground truth in the bag.

- Recover relative pose from the matches (essential matrix with the camera intrinsics).
- Add the second camera (stereo) and fuse IMU data for VIO.